＊研究論文＊

Intensity-Based Bayesian Framework for Image Reconstruction from Sparse Projection Data

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Abstract

This paper presents a Bayesian framework for iterative image reconstruction from projection data measured over a limited number of views. The classical Nyquist sampling rule yields the minimum number of projection views required for accurate reconstruction. However, challenges exist in many medical and industrial imaging applications in which the projection data is undersampled. Classical analytical reconstruction methods such as FBP are not a good choice for use in such cases because the data undersampling in the angular range introduces aliasing and streak artifacts that degrade lesion detectability. In this paper, we propose a Bayesian framework for ML-EM-based iterative reconstruction methods that incorporates *a priori* knowledge obtained from expected intensity information. The proposed framework is based on the fact that, in tomographic imaging, it is often possible to expect a set of intensity values of the reconstructed object with relatively high accuracy. The image reconstruction cost function is modified to include the ℓ_1 norm distance to the *a priori* known information. The proposed method has the potential to regularize the solution to reduce artifacts without missing lesions that cannot be expected from the *a priori* information. Numerical studies showed a significant improvement in image quality and lesion detectability under the condition of highly undersampled projection data.

Key words: Tomography, Image reconstruction, Sparse projection data, Bayesian method, Compressed sensing

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1. Introduction

In Computed Tomography (CT), projection data of the target object is measured over a set of many view angles. The number of acquired projection views has a large influence to accuracy and stability of the image reconstruction. In many imaging applications, the projection data is available only over a limited number of views. The term *limited view projection data* (also called *sparse projection data*) refers to the case where the projection data is acquired over a small number of equally/unequally spaced angles less than the requirement by the Nyquist sampling rule〔1〕. It is well-known that image reconstruction from the sparse projection data is an underdetermined problem and the reconstructed image suffers from streak/aliasing artifacts that significantly degrade lesion detectability. **Fig. 1** shows two CT imaging configurations for the conventional full scan and sparse scan. We note that image reconstruction from the sparse projection data is an important problem in both medical and industrial imaging applications. In the following, we show examples for which the projection data becomes sparse. Electrocardiogram (ECG) gated cardiac CT is a novel imaging technique used in diagnosing coronary artery disease thanks to its remarkable ability to reduce cardiac motion artifacts〔2〕. The CT imaging is synchronized with the ECG signal representing the cardiac motion and the projection data is re-organized such that all views correspond to each ECG temporal

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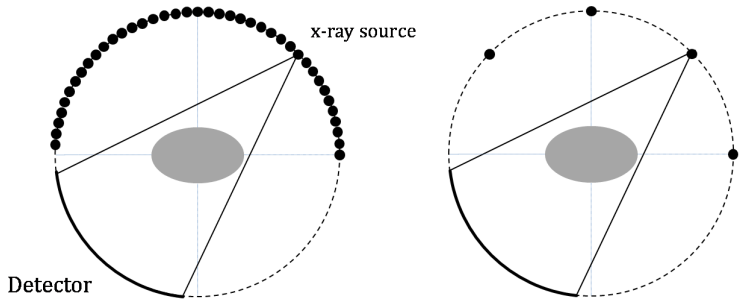

Fig. 1 Positions of x-ray source for CT data acquisition. The conventional full projection data (left) and the sparse projection data (right).

window. Even by using high-speed modern CT scanners, it is known that the re-organization of measured data still leads to a sparse projection data. Another important application of this paper is the nondestructive testing (NDT) in industry using CT techniques〔3〕. The NDT is used to examine internal cracks in various products and the sparse projection data occurs due to limitations in the scanner geometry, imaging setup and data acquisition time.

In the literature, different approaches have been investigated to reduce the streak/aliasing artifacts caused by the data undersampling. However, these approaches have not been used in commercial scanners yet. One direction to overcome this problem is to estimate missing projection data by iterative or non-iterative methods〔4, 5〕. Another direction is to incorporate *a priori* information for the scanned object into image reconstruction using Bayesian framework. The *a priori* information can be expressed in different forms, such as Gaussian smoothness prior〔6〕 and Markov random field prior〔7〕.

Recently, reconstruction of sparse signals/images is attracting attention of many research groups. Here, the term *sparse image* means that the image to be reconstructed contains a small number of non-zero pixels. The recently emerged theories of compressed sensing〔8〕demonstrate that it is possible to reconstruct accurate images using projection data measured over a small number of views fewer than the traditional requirement〔9, 10〕. The main idea is to include a distance function consisting of ℓ_1/ℓ_0 norm of the reconstructed image into the cost function for image reconstruction. Our group also applied this concept to blood vessel reconstruction〔11〕, Anatomical-MAP PET/SPECT reconstruction〔12〕and image reconstruction from truncated projection data〔13〕. Reconstruction methods based on the compressed sensing have been also developed for respiratory-gated cone-beam CT〔14〕and MRI imaging〔15〕.

In this paper, we propose a new Bayesian framework to develop iterative image reconstruction methods from the sparse projection data. In the compressed sensing framework, the image sparseness is enforced by transforming the image into some domain (e.g. gradient or wavelet) such that the majority of coefficients are approximately zeros. However, the sparsity assumption of object is not necessary in the proposed framework, as most applications in CT deal with non-sparse objects. In the following, we explain the main idea of the proposed framework. Assuming that $\vec{x}$ denotes the image vector, we represent the reconstructed image as a sum of two components; the *reference image* ($\vec{x}^R$) which includes all regions expected from the *a priori* known information (such as bones, soft tissues,..., etc) and the *target image* ($\vec{x}^T$) which includes unknown regions (such as lesions) that cannot be expected from the *a priori* information, as shown in **Fig. 2**. Medical imaging dedicated for diagnosis as well as NDT applications, in many cases, aims at discovering the existence of $\vec{x}^T$ in the scanned object and/or detecting a temporal change or an individual difference in the image. By using this image model, we consider the following two imaging scenarios. In the first scenario, we assume that $\vec{x}^R$ is *a priori* known with high accuracy. This *a priori* information can be obtained, for example, by using the image reconstructed from all measured projection data in ECG gated cardiac CT. The reference image in this case is known to be blurred due to the motion artifacts but is still valid to be used as $\vec{x}^R$. In the second scenario, we assume that both $\vec{x}^T$ and $\vec{x}^R$ are *a priori* unknown but a set of intensity values

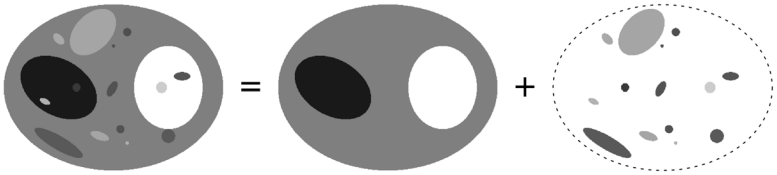

Fig. 2 The image model used in the proposed Bayesian framework. Reconstructed image $\vec{x}$ (left), reference image $\vec{x}^R$ (middle) and target image $\vec{x}^T$ (right).

$\vec{m} = (m_1, \dots, m_H)$ used to represent $\vec{x}^R$ are known in prior to image reconstruction. The term *intensity value* refers to the value of x-ray attenuation coefficient contained in the reference image $\vec{x}^R$. Each component of $\vec{m}$ corresponds to an intensity value of a single region (organ) in $\vec{x}^R$. For example, the intensity values for air, water, lung, and sternum for x-ray radiation of 70kev are known to be almost 0.0 (cm^{-1}), 1.0 (cm^{-1}), 0.26 (cm^{-1}), and 1.25 (cm^{-1}), respectively. Compared to the first scenario, the second scenario is not so difficult to be used in many CT applications. For example, in chest imaging, the intensity values of organs can be accurately expected by using a previous scan of the same patient, an image averaged for different patients or commonly known intensity values. Moreover, $\vec{m}$ can be estimated from the histogram of initial image reconstructed from the sparse projection data. We note that our group has already proposed a theoretically similar framework to the one proposed here for the first scenario. That framework was used to reconstruct PET/SPECT images with superior noise/contrast properties using template images estimated from CT/MRI〔12〕. Also, a related work to the reconstruction method proposed for the first scenario was recently published〔14〕. To the best of our knowledge, the reconstruction method proposed for the second scenario is an original contribution. This paper is organized as follows. In section 2, the proposed Bayesian framework and the associated iterative reconstruction methods are explained. In section 3, numerical studies performed to evaluate the proposed framework are shown. In section 4, conclusion and future work are described.

2. Methodology

1) Problem definition

The aim of CT imaging is to reconstruct an object from measured projection data. When iterative techniques are used for image reconstruction, this problem can be formulated as solving the linear equation $A\vec{x} = \vec{y}$ where $\vec{x} = (x_1, \dots, x_n)$ is the discrete representation of object, $A = \{a_{ij}\}$ is the $m \times n$ system matrix and $\vec{y} = (y_1, \dots, y_m)$ is the measured projection data. When the dimension of $\vec{y}$ is smaller than the dimension of $\vec{x}$ (i.e. $m < n$), this problem becomes underdetermined and reconstructing an accurate image becomes challenging. To overcome this problem, the so-called Bayesian method takes *a priori* information on the scanned object into account. The proposed Bayesian framework is based on the image model that the reconstructed image $\vec{x}$ can be represented as $\vec{x} = \vec{x}^R + \vec{x}^T$ as stated in section 1. Given $\vec{y}$ measured over a small number of views, the proposed reconstruction method is based on minimizing the following cost function.

$$f_\beta(\vec{x}) = L(\vec{x}) + \beta D(\vec{x}) \tag{1}$$

$$L(\vec{x}) = \sum_{i=1}^{m} \left[\sum_{j=1}^{n} a_{ij} x_j - y_i \log\left(\sum_{j=1}^{n} a_{ij} x_j \right) \right] \tag{2}$$

The formulation of **Eq. (1)** is well-known as maximum *a posteriori* (MAP) reconstruction, where $L(\vec{x})$ is the negative log-likelihood function, $D(\vec{x})$ is the distance function between the reconstructed image $\vec{x}$ and the available *a priori* information (described below) and β is the hyperparameter to control the strength of prior information. The distance function $D(.)$ can be formulated in various forms dependent on image properties and the nature of available *a priori* information. In this paper, we consider image reconstruction corresponding to the two scenarios described in section 1. In the first scenario (i.e. the *a priori* information takes the form of reference image $\vec{x}^R$), we propose the distance function as

$$D(\vec{x}) \equiv D_R(\vec{x}, \vec{x}^R) = \sum_{j=1}^{n} d(x_j, x_j^R). \tag{3}$$

In the second scenario, (i.e. the *a priori* information is a set of intensity values $\vec{m}$), we propose the distance function as

$$D(\vec{x}) \equiv D_I(\vec{x}, \vec{m}) = \sum_{j=1}^{n} \min_h [\omega_h d(x_j, m_h)] \tag{4}$$

The distance function in **Eq. (3)** simply measures the sum of pixel-by-pixel distance between $\vec{x}$ and $\vec{x}^R$. On the other hand, the distance function in **Eq. (4)** is defined as the sum of pixel-by-pixel minimum distance measured between each image pixel x_j and the set of *a priori* known intensity values m_h $(h = 1, \dots, H)$. The parameters $\omega_1, \dots, \omega_H$ represent weights corresponding to the components of $\vec{m}$. For example, the weights $\omega_1, \dots, \omega_H$ can be determined from the estimated intensity histogram of reference image $\vec{x}^R$ (i.e. ω_h should be large if many pixels in $\vec{x}^R$ have the intensity m_h). The single variable distance function $d(.,.)$ should be carefully selected because it has a large influence on the reconstructed image 〔12〕. The conventional selection is the least square distance (i.e. ℓ_2 norm of $\vec{x} - \vec{x}^R$), however, we have found that a good choice that guarantees the convergence of associated iterative methods and, at the same time, achieves significantly better image quality is the ℓ_1 norm defined by

$$d(a, b) = |a - b|. \tag{5}$$

In the following section, we discuss some property of the ℓ_1 norm that clarifies why ℓ_1 norm is the proper distance to be used in **Eqs. (3)** and **(4)**.

2) Power of ℓ_1 norm distance

We explain why the ℓ_1 norm behaves powerfully compared to the conventional ℓ_2 norm in finding a solution of the problem discussed above〔11, 12〕. We first note that, in the proposed image model (**Fig. 2**), $\vec{x}$ is relatively close to $\vec{x}^R$ as lesions (i.e. unexpected abnormal regions) usually arise in a small region. Therefore, the difference between $\vec{x}$ and $\vec{x}^R$ corresponding to $\vec{x}^T$ becomes a sparse image. Let us consider the simple case of a two-components image $\vec{x} = (x_1, x_2)$ and a single-component projection data $\vec{y} = (y_1)$. **Fig. 3** illustrates equi-contour lines corresponding to the ℓ_p norm with $p = 1$ and $p = 2$. The dashed line $\mathcal{L}$ represents the linear equation $a_{11}x_1 + a_{12}x_2 = y_1$. In this case, the intersection point between $\mathcal{L}$ and x_1 (or x_2) axis (centered at the reference image $\vec{x}^R$) is the sparse solution which we would like to reconstruct. From the equi-contour lines shown in **Fig. 3**, it is clear that the ℓ_1 norm distance gives a larger cost to the points far from the coordinate axes (i.e. non-sparse solutions) compared to the ℓ_2 norm. Consequently, we expect that the ℓ_1 norm distance has a stronger power in finding sparse solutions on or near the coordinate axes. This fact has been demonstrated in a few examples by our previous work〔11, 12〕. In higher-dimension problems, the difference between the ℓ_1 and ℓ_2 norms can be significant. This clarifies that, when the solution is sparse, the use of ℓ_1 norm is the method of choice. Though the ℓ_p norm with $0 < p < 1$ has a stronger power from this point of view, we preferred to use the ℓ_1 norm because the convergent iterative method is known for the ℓ_1 norm 〔16〕.

3) Iterative reconstruction methods

The main obstacle in using the ℓ_1 norm is the lack of convexity and differentiability of the cost function $f_\beta(\vec{x})$. Overcoming this drawback was made possible by using the so-called majorization-minimization strategy〔16〕. The derivation of iterative reconstruction methods to minimize $f_\beta(\vec{x})$ is based on majorizing the non-separable part of $f_\beta(\vec{x})$ by a separable quadratic function around the current iterate $\vec{x} = \vec{x}^{(k)}$ (k is the iteration number) and then minimizing the resulting cost function at each iteration. The detailed derivation using this technique can be found in 〔12〕 (we will not repeat the derivation here). The final form of the iterative methods is expressed as repeating the following two-substep procedure. The first substep is the normal image update of standard maximum likelihood expectation-maximization (ML-EM) algorithm, and the second substep is the thresholding operation which rule is determined by the *a priori* information. In the following, we outline the final form of two iterative methods associated with the two different imaging scenarios described in section 1. The main difference between the two methods

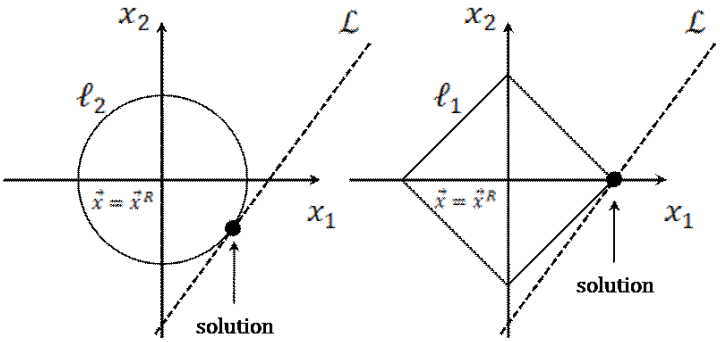

Fig. 3 Equi-contour lines of the ℓ_p norm distance with $p=2$ (left) and $p=1$ (right). $\mathcal{L}$ is the straight line representing the linear equation to be solved and the solid point represents the solution obtained by minimizing the ℓ_p norm distance.

is the nature of *a priori* information and the way to incorporate it into the cost function. The first method, R-MAP (Reference-MAP) is used in the case where the reference image $\vec{x}^R$ is available. On the other hand, the second method, I-MAP (Intensity-MAP) is used in the case where the *a priori* information is limited to only a set of intensity values $\vec{m} = (m_1, \ldots, m_H)$ arranged in ascending order (i.e. $m_1 < m_2 < \cdots < m_H$). The R-MAP and I-MAP methods are summarized as follows.

R-MAP reconstruction method

[STEP 1] (Preprocessing) Prepare the reference image $\vec{x}^R$.

[STEP 2] (Initialization) Set the initial image as $\vec{x}^{(0)} = \vec{x}^R$. Set the iteration number as $k \leftarrow 0$.

[STEP 3-1] (EM-update) Compute the image vector $\vec{p} = (p_1, p_2, \ldots, p_n)$ by

$$p_j = \frac{x_j^{(k)}}{\sum_{i=1}^{m} a_{ij}} \sum_{i=1}^{m} \frac{a_{ij} y_i}{\sum_{j'=1}^{n} a_{ij'} x_{j'}^{(k)}} . \tag{6}$$

[STEP 3-2] (Intensity thresholding) Compute the image vector $\vec{q} = (q_1, q_2, \ldots, q_n)$ by

$$q_j = \begin{cases} p_j - \delta_j & p_j > x_j^R + \delta_j \\ p_j + \delta_j & p_j < x_j^R - \delta_j \\ x_j^R & elsewhere \end{cases}, \quad \delta_j = \frac{\beta x_j}{\sum_i a_{ij}} . \tag{7}$$

[STEP 3-3] (Image update) Compute $\vec{x}^{(k+1)} = (x_1^{(k+1)}, x_2^{(k+1)}, \ldots, x_n^{(k+1)})$ by $x_j^{(k+1)} = \max\,(q_j, \epsilon)$, where $\epsilon > 0$ is a small number to ensure the positivity of pixel value (this technique is sometimes used in ML-EM-based iterative algorithms to avoid dividing by zero in **Eq. (6)**).

[STEP 4] (Convergence check) Increment the iteration number as $k \leftarrow k+1$ and go to **[STEP 3-1]** until reaching to a stopping criteria.

I-MAP reconstruction method

[STEP 1] (Preprocessing) Prepare the set of *a priori* known intensity values $\vec{m}$.

[STEP 2] (Initialization) Set the initial image as $\vec{x}^{(0)} = \varepsilon$ where $\varepsilon > 0$ is a small positive number. Set the iteration number as $k \leftarrow 0$.

[STEP 3-1] (EM-update) Compute the image vector $\vec{p} = (p_1, p_2, \ldots, p_n)$ by **Eq. (6)**.

[STEP 3-2] (Intensity thresholding) Compute the image vector $\vec{q} = (q_1, q_2, \ldots, q_n)$ by

$$q_j = \begin{cases} p_j + \delta_{j,1} & p_j < M_{j,1}^- \\ m_1 & M_{j,1}^- \le p_j \le \min\left(M_{j,1}^+, s_1\right) \\ p_j - \delta_{j,1} & M_{j,1}^+ < p_j \le s_1 \\ \vdots & \\ p_j + \delta_{j,h} & s_{h-1} < p_j < M_{j,h}^- \\ m_h & \max\left(M_{j,h}^-, s_{h-1}\right) \le p_j \le \min\left(M_{j,h}^+, s_h\right), \quad h = 2,3,\ldots,H-1 \\ p_j - \delta_{j,h} & M_{j,h}^+ < p_j \le s_h \\ \vdots & \\ p_j + \delta_{j,H} & s_{H-1} < p_j < M_{j,H}^- \\ m_H & \max\left(M_{j,H}^-, s_{H-1}\right) \le p_j \le M_{j,H}^+ \\ p_j - \delta_{j,H} & p_j > M_{j,H}^+ \end{cases} \tag{8}$$

where $\delta_{j,h} = \left(\beta x_j^{(k)} \omega_h\right) / \left(\sum_i a_{ij}\right)$, $s_h = (\omega_h m_h + \omega_{h+1} m_{h+1}) / (\omega_h + \omega_{h+1})$, $M_{j,h}^- = m_h - \delta_{j,h}$ and $M_{j,h}^+ = m_h + \delta_{j,h}$.

[STEP 3-3] (Image update) Compute $\vec{x}^{(k+1)} = (x_1^{(k+1)}, x_2^{(k+1)}, \ldots, x_n^{(k+1)})$ by $x_j^{(k+1)} = \max\ (q_j, \epsilon)$.

[STEP 4] (Convergence check) Increment the iteration number as $k \leftarrow k+1$ and go to **[STEP 3-1]** until reaching to a stopping criteria.

The thresholding function in **Eq**. **(7)** for the R-MAP method has the form of well-known soft-thresholding used in the wavelet denoising. In practical implementation, the soft-thresholding trims the image pixel value p_j in **[STEP 3-2]** toward the value of x_j^R when they are relatively close, otherwise, p_j is shifted "softly" toward x_j^R. On the other hand, the thresholding function in **Eq. (8)** for the I-MAP method can be expressed as a combination of multiple successive soft-thresholding functions. The meaning of this thresholding is as follows. If the pixel value p_j is close to any component m_h of the intensity vector $\vec{m}$ (i.e. located inside the thresholding window controlled by the parameter ω_h), p_j is trimmed to m_h. Otherwise, p_j is shifted "softly" toward the closest value among $m_1, m_2, \ldots, m_H$. The thresholding functions in both methods are designed to find the correct pixel value using the *a priori* information when the projection data is sparse. As shown later, the proposed framework significantly reduces the streak/aliasing artifacts.

3. Experimental results

1) Numerical phantom

In this simulation, we implemented ordered-subsets versions of the proposed methods (R-MAP and I-MAP) and the conventional OS-EM method. The phantom shown in **Fig. 2** was used and noise free projection data was used for image reconstruction. The image size was 512×512 (pixels) and the projection data was computed by parallel-beam geometry with 512 (radial bins) and 8/16 views (over 180°). In this phantom, the pixel intensity value inside each region is uniform. For the R-MAP method, we assumed that the accurate reference image $\vec{x}^R$ is *a priori* known with no registration errors (ideal setup). For the I-MAP method, the intensity values representing the *a priori* information were $\vec{m}$ = (0.0, 0.2, 1.0, 2.0), where the value of zero corresponds to the region outside the object (air) and the other values correspond to the three regions in the reference image $\vec{x}^R$. All the iterative methods were implemented with 300 iterations and 4 subsets. The main aim of this simulation was to evaluate detectability of unexpected regions (lesions) $\vec{x}^T$ in the reconstructed image when the projection data is sparse. The reconstructed images are shown in **Fig. 4**. In the conventional OS-EM images, the severe streak/aliasing artifacts are present and the contrast of reconstructed lesions $\vec{x}^T$ is low, which reduces possibility of detecting the lesions. However, a significant improvement in the lesion detectability was achieved in both the R-MAP and I-MAP methods. By using only 16 projection views, most of lesions with different size, contrast, and topology were recognized. As described in 〔12〕, the R-MAP method, which uses the reference image $\vec{x}^R$, is sensitive to registration errors between $\vec{x}$ and $\vec{x}^R$. However, this is not the case for the I-MAP method. Moreover, as explained above, the required *a priori* information for the I-MAP method is significantly smaller than that for the R-MAP method.

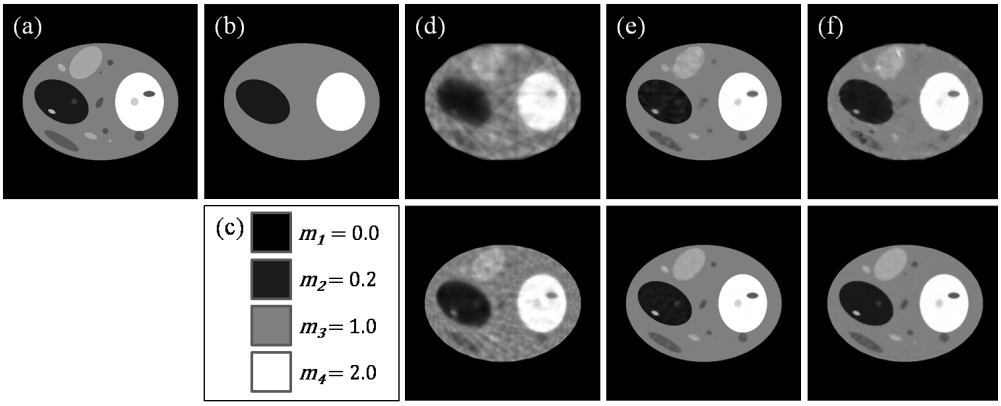

Fig. 4 (a) Numerical phantom, (b) the reference image $\vec{x}^R$ for the R-MAP method and (c) intensity values $\vec{m}$ for the I-MAP method. Reconstructed images using (d) the conventional OS-EM, (e) R-MAP and (f) I-MAP methods (8 views in top row and 16 views in bottom row). The gray scale is [0.0, 2.0].

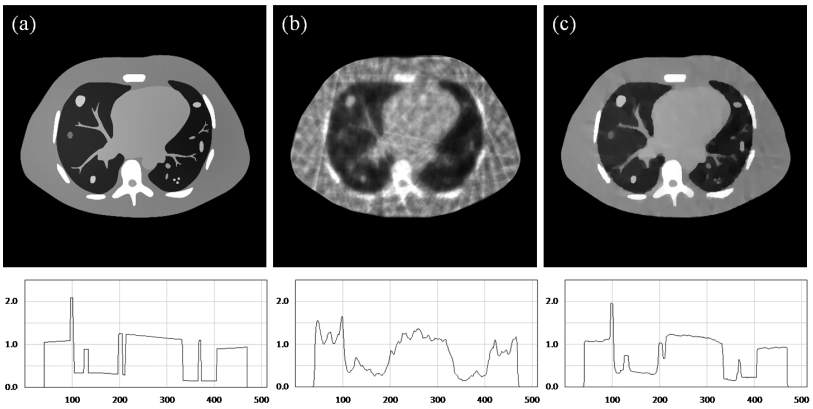

Fig. 5 (a) Chest phantom. Images (top) and central horizontal profiles (bottom) reconstructed from 16 projection views using (b) OS-EM and (c) I-MAP methods. The gray scale is [0.0, 2.0].

2）Realistic chest phantom

In clinical CT applications, the attenuation coefficient within the same organ is not absolutely uniform in general. In this study, a realistic chest phantom was used to evaluate the I-MAP method. The phantom represents a slice of human thorax and the intensity value within each region (organ) is non-uniform as shown by the profile plot in **Fig. 5(a)**. A 2D Gaussian fluctuation function was used to generate this phantom from the original segmented chest image. The chest phantom contains lung lesions with different topology, size and contrast together with the blood-vessel structures as shown in **Fig. 5(a)**. To handle the effect of intensity variation within the same region, the distance function in **Eq. (5)** was modified as

$$d(a,b) \equiv \tilde{d}(a,b,\Delta) = \begin{cases} a-b-\Delta & (a-b>\Delta) \\ b-a-\Delta & (a-b<-\Delta) \\ 0 & (otherwise) \end{cases}, \tag{9}$$

where Δ is the parameter empirically determined by the degree of intensity fluctuation within the same region. The derivation of the corresponding thresholding function in this case is direct and is omitted. The *a priori* information for the I-MAP method $\vec{m}$ = (0.0, 0.26, 1.0, 1.2, 2.0) represents values of attenuation coefficients corresponding to air, lungs, soft tissues, heart and bones. $\vec{m}$ was computed as average values of each region. In this study, the image model in **Fig. 2** was used under the assumption that the reference image $\vec{x}^R$ contains normal thorax structures (i.e. lungs, soft-tissue, heart and bones), but does not contain the other unexpected structures. On the other hand, the tar-

get image $\vec{x}^T$ contains the unexpected structures such as lesions (with different topology, size and contrast) and blood-vessels located inside the lungs. Therefore, the purpose of image reconstruction here was to improve detectability (contrast and spatial resolution) of the lesions and blood-vessels $\vec{x}^T$ in the reconstructed image when the projection data is sparse. Of course, in the I-MAP method, only the intensity values constructing the normal structures were used instead of the reference image $\vec{x}^R$ itself. The projection data was computed over 16 views and other simulation setups were similar to those in the previous study. The reconstructed images shown in **Fig. 5** demonstrate that the conventional OS-EM reconstruction suffers from severe streak/aliasing artifacts and low-contrast in the lung lesions so that lesion detectability is poor. On the other hand, by using the I-MAP, most of lesions can be recognized correctly by reducing the image artifacts and recover the contrast of lesions and blood vessels. The proposed I-MAP method still works well even if intensity value within the same region is slightly non-uniform.

4. Conclusions

We proposed a Bayesian framework to develop iterative image reconstruction methods in CT when the projection data is sparse. The proposed framework is appropriate to reduce streak/aliasing artifacts by using *a priori* information on intensity values of scanned object. The main contribution of this paper was to reduce the required *a prior*i information for image improvement to only a set of intensity values, whereas the previous similar methods assumed that the reference image is accurately known. The cost function for image reconstruction was designed by adding a distance function based on the ℓ_1 norm, which compares each pixel value of reconstructed image to the *a priori* known intensity values. We think that the *a priori* information on the intensity values is relatively easy to obtain in many clinical CT imaging situations and, therefore, the proposed framework sounds practical. The reconstructed images in the numerical studies show that the proposed framework can significantly reduce image artifacts and preserve image details unexpected from the *a priori* knowledge even if the intensity values within the same region are non-uniform. Applying the proposed framework to real patient data is currently under investigation. Future work may include investigating possible approaches to obtain the *a priori* information in various imaging situations and extending the proposed method to the limited angle and truncation problems.

Finally, we remark that the proposed Bayesian framework is general such that it has a potential to be used in the other limited-data problems. However, we expect that its validity and power strongly depend on the type of image artifacts to be eliminated by the use of *a priori* knowledge. Since such experimental evaluation must be done dependent on the type of limited-data, as a first step, this paper limited the focus-of-attention to the case of "sparse projection data", just due to the fact that it is simplest among all the limited-data problems. We will publish experimental results for the other limited-data problems in separate publications〔13〕.

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疎な投影データからの画像再構成のための濃度情報に基づくベイズ推定の枠組み

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要旨：本論文では，少数方向投影データからの逐次近似画像再構成のためのベイズ推定の新しい枠組みを提案する．正確な画像再構成に必要な投影データの方向数はナイキスト間隔により与えられるが，医療用や工業計測用の CT では少数方向の投影データしか計測できない状況が起きる．このような場合，FBP 法などの解析的画像再構成法では画像にストリーク状のエリアシングアーティファクトが生じ，病変検出能が低下する．本論文では，画像の濃度値に関する事前情報を ML-EM 法に基づく逐次近似法に組み入れて画質を改善するベイズ推定の枠組みを提案する．提案手法は，多くの CT イメージングの状況において，画像にどのような濃度値の物体が含まれているかは比較的高い精度で予想できる事実に基づいている．提案手法では，画像再構成の評価関数に事前情報と再構成画像の L1 ノルム距離を組み入れ，これにより事前情報から外れた病変を保存したままストリークアーティファクトを低減できる．数値ファントムと実 CT 画像から作成したファントムを用いたシミュレーション実験により，提案手法により少数方向の投影データから画質と病変検出能を大幅に改善した画像が得られることを示す．

キーワード：トモグラフィ，画像再構成，疎な投影データ，ベイズ推定，圧縮センシング
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